

Deploying models in production

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Disclaimer

The views and opinions expressed in this presentation are those of the author and do not necessarily reflect the position of the organization of which he belongs.

Intro

- ▶ Deploying a model in production is a complex task that requires a deep understanding of the model, the data and the infrastructure.
- ▶ Actuaries were traditionally involved in the first two aspects, but the third one is becoming more and more important.
- ▶ Modern *Python frameworks* make the third aspect *easier*, at least when presenting a POC to the stakeholders.

MLOps: Enhancing the ML Model Lifecycle

- ▶ **Definition:** MLOps integrates practices to automate and manage the entire machine learning model lifecycle, improving deployment efficiency and reliability in production.
- ▶ **Why It's Needed:** Bridges the gap between development and production, enabling better collaboration between data scientists and engineers for models that are more performing, scalable, and maintainable.
- ▶ **Key Benefits:** Unified workflow reduces errors, accelerates deployment, and provides continuous monitoring to keep models up-to-date and accurate.

MLOps: instruments

- ▶ **Model training:** Python pipelines, requirements files, MLflow
- ▶ **Version control:** Git, GitHub, GitLab
- ▶ **Model deployment:** FastAPI, Streamlit, Docker

Project presentation

- ▶ **Objective:** Deploying an insurance quote model in production
- ▶ **Tools:**
 - ▶ **Python pipelines:** to train the frequency, severity and pure premium models
 - ▶ **Git:** to version control the code
 - ▶ **Docker:** to create a container with the models
 - ▶ **FastAPI:** to deploy the models
 - ▶ **Streamlit:** to create a user interface

General recommendations

AI tools

- ▶ Code assistant tools are becoming increasingly widespread in actuarial and data science workflows, significantly improving productivity and accelerating development cycles.
- ▶ These tools can support developers in writing boilerplate code, generating documentation, debugging, and exploring alternative implementations.
- ▶ However, AI-generated code should never be accepted blindly. Outputs must always be critically reviewed, validated, and tested before being used in production environments.

- ▶ Particular attention should be paid to:
 - ▶ hidden logical errors,
 - ▶ inefficient implementations,
 - ▶ security vulnerabilities,
 - ▶ incorrect assumptions on data structures or business rules,
 - ▶ and hallucinated libraries or functions.
- ▶ In actuarial contexts, domain expertise remains essential: the final responsibility for the correctness, interpretability, and regulatory compliance of the model always lies with the practitioner.
- ▶ AI tools should therefore be considered as productivity enhancers rather than substitutes for actuarial judgment and software engineering best practices.

Data exploration and understanding

- ▶ Before training or deploying any predictive model, it is fundamental to thoroughly explore and understand the underlying data.
- ▶ Univariate and bivariate exploratory analyses are essential to:
 - ▶ understand the phenomenon being modeled,
 - ▶ identify data quality issues,
 - ▶ detect anomalies and outliers,
 - ▶ verify variable distributions,
 - ▶ assess missing values,
 - ▶ and identify potential relationships between predictors and targets.
- ▶ Exploratory analysis also helps identify:
 - ▶ incorrect formats or inconsistent encodings,
 - ▶ data leakage risks,
 - ▶ unstable variables,
 - ▶ highly imbalanced targets,
 - ▶ and unexpected correlations.

- ▶ Visualization techniques such as histograms, boxplots, scatter plots, correlation matrices, and grouped summaries often provide insights that are difficult to detect through automated pipelines alone.
- ▶ A good understanding of the data generation process is frequently more valuable than experimenting with increasingly complex models.

Production mindset

- ▶ A model that performs well in a notebook is not necessarily ready for production.
- ▶ Production-ready actuarial models should emphasize:
 - ▶ reproducibility,
 - ▶ monitoring,
 - ▶ explainability,
 - ▶ versioning,
 - ▶ robustness,
 - ▶ and maintainability.
- ▶ Whenever possible:
 - ▶ separate training and inference pipelines,
 - ▶ track model versions and datasets,
 - ▶ log predictions and input data,
 - ▶ and monitor performance drift over time.
- ▶ Simpler and interpretable models are often preferable to highly complex solutions if they provide comparable predictive performance.

Communication and documentation

- ▶ Clear documentation is a key component of production systems.
- ▶ Assumptions, limitations, preprocessing steps, feature engineering choices, and validation methodologies should be explicitly documented.
- ▶ Effective communication between actuaries, data scientists, software engineers, and business stakeholders is critical to ensure that models are correctly understood and properly integrated into operational processes.

Python pipelines

Python set up

Why Python

- ▶ It is the de-facto standard for ML
- ▶ Rich ML libraries ecosystem as well as production and deployment tools
- ▶ Well known by Computer Scientist and Data Engineers



Figure 1: Python Logo

Python environment

- ▶ It is always advisable to create an environment in the project working directory
- ▶ The environment allows to set up the necessary libraries and tools to make the workflow reproducible and isolated
- ▶ Tools like *uv* and *conda* make easier the creation of python environment

Python pipelines

Structure

- ▶ Ingests the data, and clean
- ▶ Fits the models, saves them
- ▶ Assess the performance of the models, uses MLflow to log the results and artifacts

Running the pipeline

- ▶ see the `main.py` file
- ▶ the directory `steps` contains the steps of the pipeline
- ▶ It can be run as `python main.py`

Requirements

- ▶ Python packages specified in a `requirements.txt` file
- ▶ The `requirements.txt` file is used to create a virtual environment
- ▶ The Dockerfile uses the virtual environment to create a container

MLflow

- ▶ MLflow (Alla and Adari 2020) is an open source platform for managing the end-to-end machine learning lifecycle.
- ▶ It is used to log the results and artifacts of the models, that can be inspected in the MLFlow UI at <http://localhost:5000> after running the pipeline
- ▶ It is organized in experiments and runs

Streamlit walkthrough

Why streamlit?

- ▶ Streamlit (Raghavendra 2022) is an open-source app framework for Machine Learning and Data Science projects.
- ▶ It creates a user interface for the models
- ▶ It is easy to use and to deploy

Streamlit code

- ▶ the `st.` functions are used to create the user interface
- ▶ the session state is used to track user's choices, e.g. button clicks
- ▶ the models are loaded, cached and used to make predictions

Running the app

- ▶ The app is run with `streamlit run quote-page.py`
- ▶ An initial section allows to input the policyholder's data (pre-defined values exists)
- ▶ A button click will send the data to the predict pipeline and show the results

FastAPI Walkthrough

Why FastAPI?

- ▶ FastAPI (Lathkar 2023) is a modern framework for building APIs, ideal for deploying machine learning models.
- ▶ It enables exposing models as a REST API, allowing other applications to use them, for example, via Python's `requests` library.
- ▶ It uses Python type hints to validate the input and output of the API, improving code readability and maintainability.

Structure of a FastAPI App

- ▶ The application is defined in a Python file, typically named `app.py`.
- ▶ The app runs with the ASGI server `uvicorn`, allowing FastAPI to handle asynchronous requests, which increases scalability and speed.
- ▶ The app structure includes a startup event for loading models, endpoints for handling requests, and dependencies to manage model usage.

Key Components of the Insurance Prediction API

1. Input Schema

- ▶ The input is structured using the `Insured` model, a class that describes the characteristics of the insured person, such as vehicle power (`VehPower`), vehicle age (`VehAge`), driver age (`DrivAge`), and other attributes relevant for calculating insurance premiums.
- ▶ Each field includes a title, description, and validity limits using `Field` from `pydantic`. This ensures input data consistency and completeness.

2. Output Schema

- ▶ The output is managed by the `PredictionResponse` class, defining the key variables predicted by the model: frequency (`Frequency`), severity (`Severity`), and pure premium (`Pure_Premium`).
- ▶ The `PredictionResponse` class ensures that the API always returns a standard format, simplifying integration with other applications.

3. Startup Event

- ▶ A FastAPI startup event, defined with `@app.on_event("startup")`, loads the CatBoost models from the `models` folder. This prevents reloading the models every time a request is made, improving efficiency.
- ▶ In this example, `model_freq` and `model_sev` are, respectively, the frequency and severity prediction models.

4. Prediction Endpoint

- ▶ The primary endpoint of the application is `/predict/`, which receives the insured person's data and returns a prediction.
- ▶ Within the endpoint, the `Frequency` model and `Severity` model are used to calculate the pure premium:
 - ▶ **Frequency** (`Frequency`): predicts the number of claims.
 - ▶ **Severity** (`Severity`): predicts the average cost of claims.
 - ▶ **Pure Premium** (`Pure_Premium`): calculated as `Frequency * Severity`, representing the expected premium for the customer.
- ▶ The endpoint uses FastAPI's `Depends` to load the model dependencies and then uses `Predictor` to perform predictions.

FastAPI in Action

- ▶ The API can be run locally using the command `uvicorn app:app --reload`, enabling rapid development and testing.
- ▶ Once the application is online, you can interact with the prediction endpoint using tools like `curl` or directly from FastAPI's interactive documentation available at `/docs`.

HTTP Status Codes to Discuss

- ▶ **2xx (Success):** request processed correctly.
 - ▶ 200 OK: successful /predict/, /health, /ready.
- ▶ **4xx (Client errors):** input or request issue.
 - ▶ 400 Bad Request: malformed request.
 - ▶ 401 Unauthorized: authentication missing/invalid.
 - ▶ 403 Forbidden: authenticated but not allowed.
 - ▶ 404 Not Found: endpoint does not exist.
 - ▶ 422 Unprocessable Entity: validation failed (common with Pydantic).
- ▶ **5xx (Server errors):** backend issue.
 - ▶ 500 Internal Server Error: unhandled exception in API logic.
 - ▶ 502 Bad Gateway: upstream/proxy cannot reach a healthy app.
 - ▶ 503 Service Unavailable: service running but not ready (e.g., models not loaded).
 - ▶ 504 Gateway Timeout: upstream did not respond in time.

Docker

Why Docker?

- ▶ Docker (Jangla 2018) is a platform for developing, shipping, and running applications.
- ▶ It allows you to create containers with models and dependencies, ensuring a consistent and isolated environment.
- ▶ Commonly used to deploy machine learning models in production.

Dockerfile Overview

- ▶ The Dockerfile defines steps to create a container for our FastAPI application, including:
 - ▶ Loading a base Python image
 - ▶ Copying models, code, and dependencies
 - ▶ Installing packages from `requirements.txt`
 - ▶ Starting the FastAPI app

Docker commands

- ▶ `docker build -t deployer .` to build the image
- ▶ `docker run -d --name deployer -p 8080:8080 deployer:latest` to run the container that exposes the app on port 8000
- ▶ `docker stop` `deployer` to stop the container

References I

Alla S, Adari SK (2020) Introduction to MLFlow. Apress, pp 125–227

Jangla K (2018) Docker. Apress, pp 9–17

Lathkar M (2023) Introduction to FastAPI. Apress, pp 1–28

Raghavendra S (2022) Introduction to Streamlit. Apress, pp 1–15